INCH-POUND

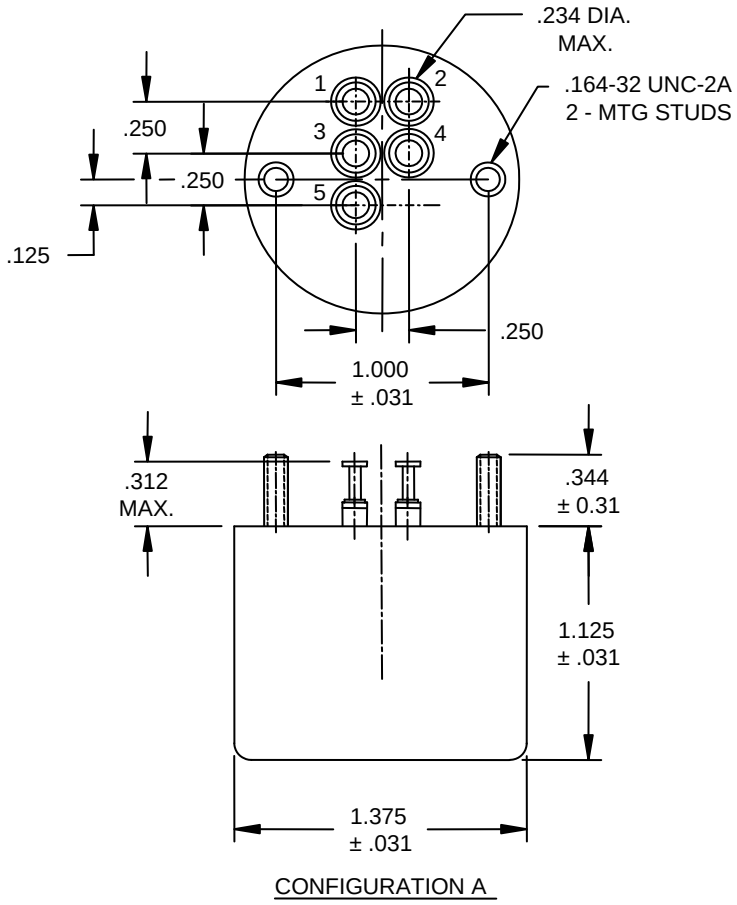
MIL-PRF-27/56E
11 December 2009
SUPERSEDING
MIL-PRF-27/56D
6 May 2005

PERFORMANCE SPECIFICATION SHEET

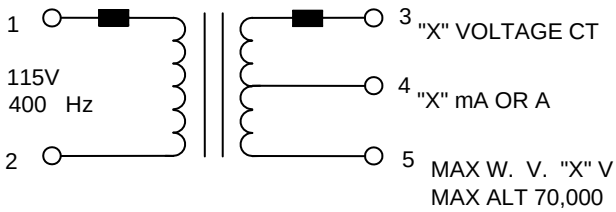
TRANSFORMERS, POWER, 25 VOLT AMPERES, 400 HZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformers described herein shall consist of this specification and MIL-PRF-27.



CONFIGURATION A



(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING

Inches	mm
.031	0.79
.125	3.18
.164	4.17
.234	5.94
.250	6.35
.312	7.92
.344	8.74
1.000	25.40
1.125	28.58
1.375	34.92

FIGURE 1. Dimensions and configuration.

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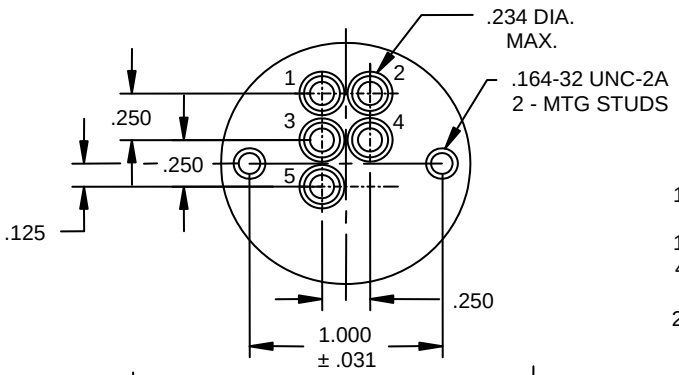
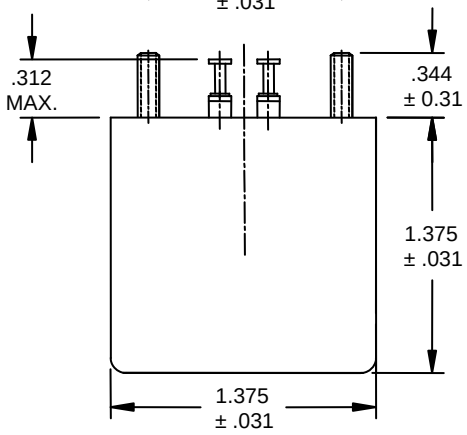
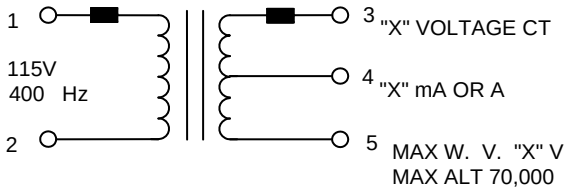




CONFIGURATION B



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.344	8.74
1.000	25.40
1.375	34.92

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the side of the case.
5. Terminals are color coded for permanent identification in addition to terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 4.95 ounces, maximum.

Duty cycle: Continuous.

Case: Metal enclosed (configuration A) or epoxy fiberglass enclosed (configuration B).

Material: Steel (configuration A) or epoxy fiberglass (configuration B).

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Feed thru.

Terminal height: .312 inch (7.92 mm), maximum.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 5 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 25 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 88 percent minimum at rated load.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{Voltage\ (no\ load) - Voltage\ (rated\ load)}{Voltage\ (rated\ load)} \quad x \quad 100$

Shall not exceed 7 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 45°C, maximum with 115 volts, 400 Hz across (1-2) at ambient temperature of 85°C.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.

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VERIFICATION:

Extent of qualification:

Qualification testing and approval to M27/49-44 A or B, M27/51-44 A or B, M27/53-44 A or B, M27/56-44 and M27/66-27 shall be sufficient to grant qualification approval to MIL-PRF-27/48 through MIL-PRF-27/66 inclusive, all parts.

Qualification testing and approval to M27/56-44 shall be sufficient to grant qualification approval to M27/56-01 through M27/56-44.

Part or Identifying Number (PIN) example:

M	27/56	-01
MIL specification item	Specification sheet no.	Dash no.

TABLE I. Electrical ratings.

Dash number	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)	Configuration
01	6	4.167	325	A
02	6.3	3.968	325	A
03	7	3.571	325	A
04	7.5	3.333	325	A
05	8	3.125	325	A
06	9	2.778	325	A
07	10	2.500	325	A
08	11	2.273	325	A
09	12	2.083	325	A
10	12.6	1.984	325	A
11	13	1.923	325	A
12	15	1.667	325	A
13	16	1.563	325	A
14	18	1.389	325	A
15	20	1.250	325	A
16	22	1.136	325	A
17	24	1.042	325	A
18	26	0.962	325	A
19	28	0.893	325	A
20	30	0.833	325	A
21	33	0.758	325	A
22	36	0.694	325	A
23	39	0.641	325	A
24	43	0.581	325	A
25	47	0.532	325	A
26	51	0.490	325	A
27	56	0.446	325	A
28	62	0.403	325	A
29	68	0.367	325	A
30	75	0.333	325	A
31	82	0.305	325	A
32	91	0.275	325	A
33	100	0.250	325	A
34	110	0.227	325	A
35	115	0.218	325	A
36	120	0.208	325	A
37	130	0.192	325	A
38	150	0.166	325	A
39	180	0.139	325	A
40	200	0.125	440	A
41	220	0.113	440	A
42	240	0.104	440	A
43	270	0.093	580	A
44	300	0.083	580	A

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TABLE I. Electrical ratings - Continued.

Dash number	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)	Configuration
45	6	4.167	325	B
46	6.3	3.968	325	B
47	7	3.571	325	B
48	7.5	3.333	325	B
49	8	3.125	325	B
50	9	2.778	325	B
51	10	2.500	325	B
52	11	2.273	325	B
53	12	2.083	325	B
54	12.6	1.984	325	B
55	13	1.923	325	B
56	15	1.667	325	B
57	16	1.563	325	B
58	18	1.389	325	B
59	20	1.250	325	B
60	22	1.136	325	B
61	24	1.042	325	B
62	26	0.962	325	B
63	28	0.893	325	B
64	30	0.833	325	B
65	33	0.758	325	B
66	36	0.694	325	B
67	39	0.641	325	B
68	43	0.581	325	B
69	47	0.532	325	B
70	51	0.490	325	B
71	56	0.446	325	B
72	62	0.403	325	B
73	68	0.367	325	B
74	75	0.333	325	B
75	82	0.305	325	B
76	91	0.275	325	B
77	100	0.250	325	B
78	110	0.227	325	B
79	115	0.218	325	B
80	120	0.208	325	B
81	130	0.192	325	B
82	150	0.166	325	B
83	180	0.139	325	B
84	200	0.125	440	B
85	220	0.113	440	B
86	240	0.104	440	B
87	270	0.093	580	B
88	300	0.083	580	B

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Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-PRF-27/48	MIL-PRF-27/49	MIL-PRF-27/50	MIL-PRF-27/51	MIL-PRF-27/52
MIL-PRF-27/53	MIL-PRF-27/54	MIL-PRF-27/55	MIL-PRF-27/57	MIL-PRF-27/58
MIL-PRF-27/59	MIL-PRF-27/60	MIL-PRF-27/61	MIL-PRF-27/62	MIL-PRF-27/63
MIL-PRF-27/64	MIL-PRF-27/65	MIL-PRF-27/66	MIL-STD-202	

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Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Review activities:
Army – CR4, MI
Navy – AS, MC, OS
Air Force – 19, 99

Preparing activity:
DLA - CC

(Project 5950-2010-001)

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